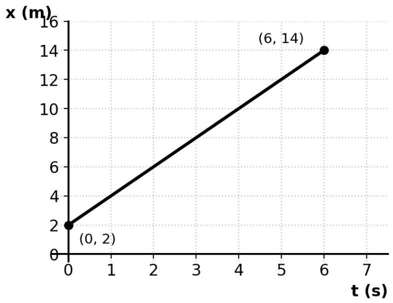
KINEMATICS GRAPHS WORKSHEET

Class 11 · JEE Foundation

Q1.

The position–time graph of a particle is shown below.



(a) Find the velocity of the particle.

(b) Find its position at t = 3 s.

Q2.

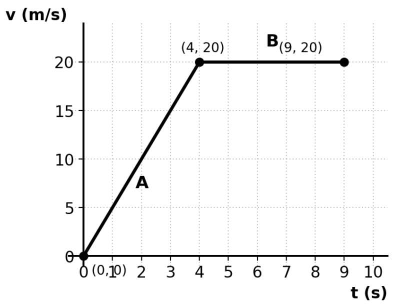
A particle moves 20 m in 4 s at constant velocity, then stays at rest for the next 3 s.

(a) Find the velocity during the first 4 s.

(b) Find the total displacement over the 7 s.

Q3.

The velocity–time graph of a scooter is shown below, with regions A and B marked.



(a) Find the acceleration in regions A and B.

(b) Find the total displacement over the full 9 s.

Q4.

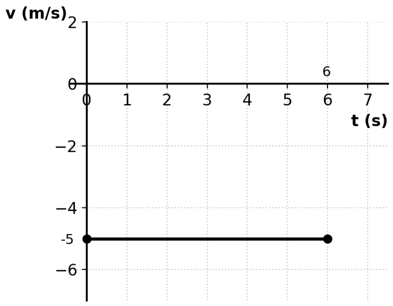
A car's velocity falls uniformly from 24 m/s to 0 in 6 s.

(a) Find the acceleration of the car.

(b) Find the distance it covers in these 6 s.

Q5.

The velocity–time graph of a particle is shown below.



(a) Find the displacement over the 6 s shown.

(b) Find the distance travelled over the same 6 s.

Q6.

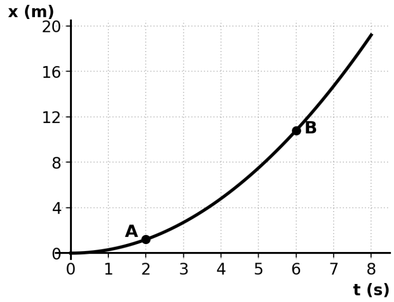
A scooter moves at a constant 10 m/s for 3 s, then stops instantly and stays at rest for the next 3 s.

(a) Find the total displacement over the 6 s.

(b) Find the average velocity over the 6 s.

Q7.

The position–time graph of a particle is shown below, with points A and B marked.

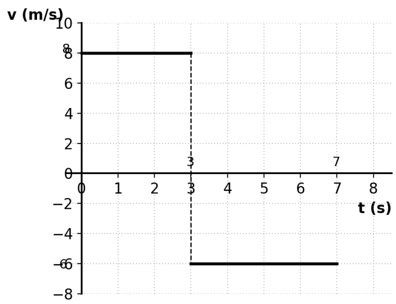


(a) At which point, A or B, is the particle moving faster?

(b) Between A and B, is the speed increasing or decreasing?

Q8.

A particle's velocity–time graph is shown below: +8 m/s for the first 3 s, then –6 m/s for the next 4 s.



(a) Find the total displacement of the particle.

(b) Find the total distance travelled.

Q9.

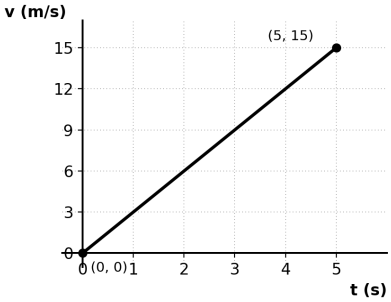
A train starts from rest and accelerates uniformly to 30 m/s in 6 s, runs at 30 m/s for the next 10 s, then decelerates uniformly to rest in 5 s.

(a) Find the acceleration in the first and last phases.

(b) Find the total distance covered by the train.

Q10.

A particle starts from the origin with the velocity–time graph shown below (a straight line through the origin).

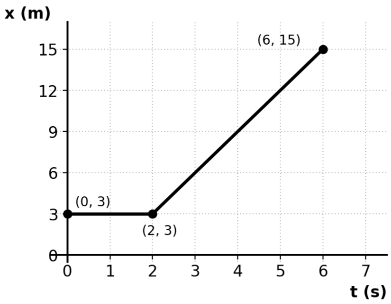


(a) Find the shape and value of the acceleration–time graph.

(b) Find displacement x as a function of t, and state the shape of the resulting x–t graph.

Q11.

A particle's position is plotted against time below. It stays still for the first 2 s, then starts moving.



(a) Find the velocity during 0–2 s and during 2–6 s.

(b) Find the average velocity over the full 6 s.

Q12.

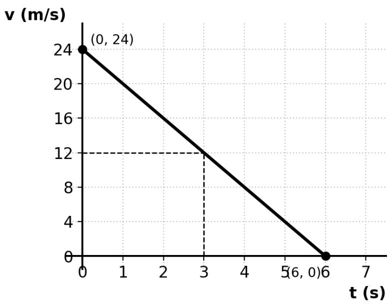
A ball is thrown straight up with a speed of 20 m/s and returns to the thrower's hand 4 s later.

(a) Find its velocity at the highest point.

(b) Find its acceleration on the way up and on the way down.

Q13.

The velocity–time graph of a car is shown below.



(a) Find the acceleration of the car.

(b) Find the velocity of the car at t = 3 s.

Q14.

A particle moves at +7 m/s for 3 s, then at –3 m/s for the next 5 s.

(a) Find the displacement of the particle.

(b) Find the distance travelled.

Q15.

A particle starts from rest with a constant acceleration of 3 m/s², maintained for 4 s.

(a) Find the change in velocity over these 4 s.

(b) Find the final velocity of the particle.

Q16.

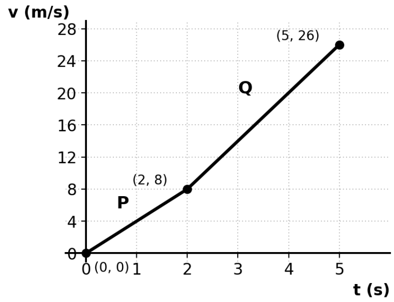
A car's speed increases uniformly from 0 to 15 m/s in 5 s.

(a) Find the acceleration of the car.

(b) Find the distance it covers in these 5 s.

Q17.

The velocity–time graph of a particle is shown below, with points P and Q marked.



(a) Find the acceleration just before P and just after P.

(b) In which part, before or after P, is the acceleration greater?

Q18.

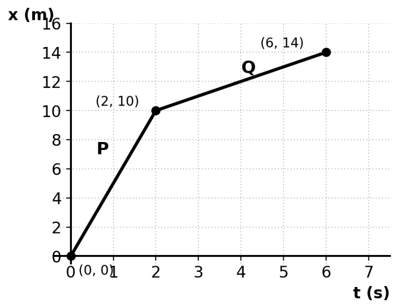
A particle stays at rest for 3 s, then moves with constant acceleration and reaches 12 m/s at t = 7 s.

(a) Find its acceleration during 3–7 s.

(b) State the shape of its x–t graph during 3–7 s.

Q19.

The position–time graph of a particle is shown below, with points P and Q marked.



(a) Find the velocity in the segment before P–Q and after.

(b) Which segment shows the higher speed?

Q20.

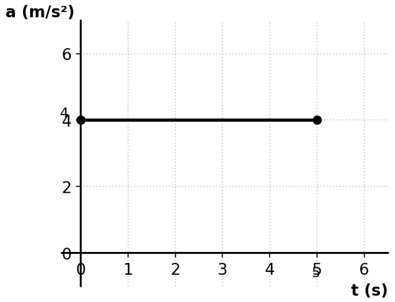
A lift moves up at a constant 3 m/s for 8 s, then decelerates uniformly to rest in the next 2 s.

(a) Find the deceleration of the lift.

(b) Find the total distance travelled by the lift.

Q21.

The acceleration–time graph of a particle is shown below. It starts from rest.



(a) Find the change in velocity over the 5 s shown.
(b) Find the final velocity of the particle.

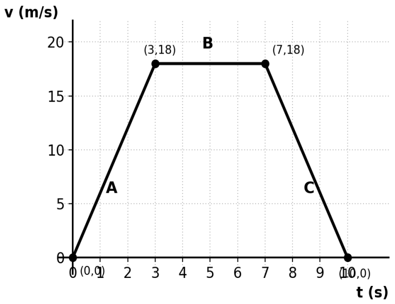
Q22.

A particle moves with a constant velocity of –8 m/s for 5 s.

(a) Find its displacement over these 5 s.
(b) State the direction of motion of the particle.

Q23.

A bus accelerates, cruises, then decelerates, as shown below, with phases A, B, C marked.



(a) Find the acceleration in phases A and C.
(b) Find the total displacement over the full 10 s.

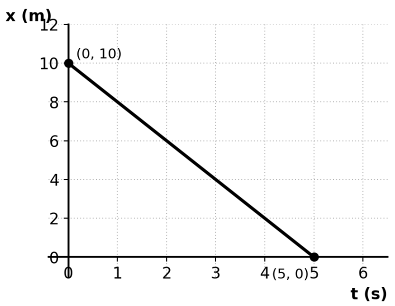
Q24.

A stone is dropped from rest and falls for 3 s. Take $g = 10$ m/s².

(a) Find its velocity at $t = 3$ s.
(b) Find the distance it falls in these 3 s.

Q25.

The position–time graph of a particle is shown below.



(a) Find the velocity of the particle, with sign.
(b) State the direction of motion.

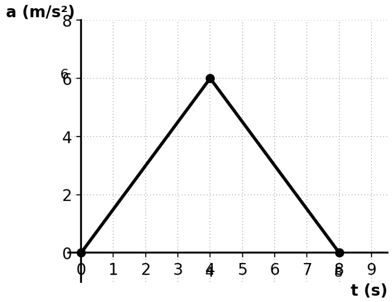
Q26.

A cyclist's velocity increases uniformly from 4 m/s to 16 m/s in 4 s.

(a) Find the acceleration of the cyclist.
(b) Find the distance covered in these 4 s.

Q27.

The acceleration–time graph of a particle is shown below. It starts from rest.



(a) Find the change in velocity during 0–4 s.
(b) Find the final velocity of the particle at $t = 8$ s.

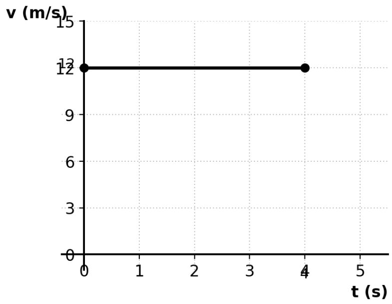
Q28.

A particle moving at 5 m/s comes to rest in 2 s under uniform deceleration.

(a) Find the deceleration of the particle.
(b) Find the distance it travels before stopping.

Q29.

The velocity–time graph of a particle is shown below.



(a) State the shape of the corresponding x–t graph.
(b) State the shape of the corresponding a–t graph.

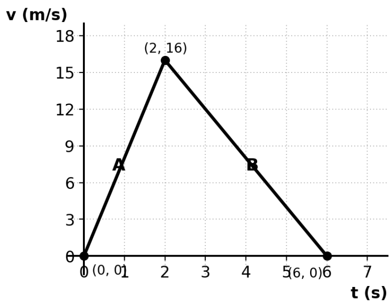
Q30.

A ball is dropped from a height and hits the ground 4 s later. Take $g = 10$ m/s².

(a) Find its velocity just before hitting the ground.
(b) Find the height from which it was dropped.

Q31.

The velocity–time graph of a particle is shown below, with phases A and B marked.



(a) Find the acceleration in phases A and B.
(b) Find the total displacement over the 6 s.

Q32.

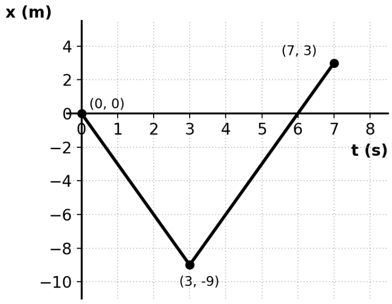
A car moves at 20 m/s for 5 s, then instantly reverses direction and moves at 15 m/s for the next 4 s.

(a) Find the net displacement of the car.

(b) Find the total distance travelled.

Q33.

The position–time graph of a particle is shown below.



(a) Find the velocity in each segment.

(b) Find the net displacement and the total distance over the 7 s.

Q34.

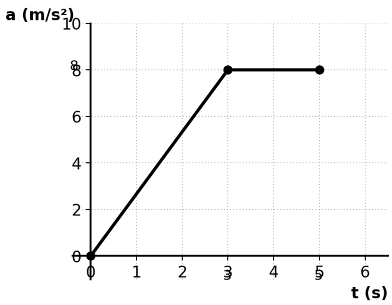
A ball is thrown vertically upward with a speed of 15 m/s. Take g = 10 m/s².

(a) Find the time it takes to reach the highest point.

(b) Find the maximum height reached.

Q35.

The acceleration–time graph of a particle is shown below. It starts from rest.



(a) Find the change in velocity during 0–3 s.

(b) Find the final velocity of the particle at t = 5 s.

Q36.

A scooter accelerates uniformly from 6 m/s to 18 m/s in 4 s, then holds 18 m/s for the next 5 s.

(a) Find the acceleration during the first 4 s.

(b) Find the total distance covered over the 9 s.

Q37.

A particle moves at +10 m/s for 2 s, stays at rest for the next 3 s, then moves at –5 m/s for 2 s.

(a) Find the total displacement of the particle.

(b) Find the total distance travelled.

Q38.

A train decelerates uniformly from 40 m/s to 10 m/s in 5 s.

(a) Find the acceleration of the train.

(b) Find the distance it covers in these 5 s.

ANSWER KEY

1. v = 2 m/s; x(3 s) = 8 m
2. v = 5 m/s; displacement = 20 m
3. a(A) = 5 m/s², a(B) = 0; displacement = 140 m
4. a = –4 m/s²; distance = 72 m
5. displacement = –30 m; distance = 30 m
6. displacement = 30 m; average velocity = 5 m/s
7. B is faster; speed is increasing throughout
8. displacement = 0 m; distance = 48 m
9. a(1st phase) = 5 m/s², a(last phase) = –6 m/s²; distance = 465 m
10. a–t: flat, a = 3 m/s²; x = 1.5t², x–t is a parabola
11. v = 0 (0–2 s), v = 3 m/s (2–6 s); average velocity = 2 m/s
12. velocity at top = 0; a = –10 m/s² throughout (up and down)
13. a = –4 m/s²; v(3 s) = 12 m/s
14. displacement = 6 m; distance = 36 m
15. Δv = 12 m/s; final velocity = 12 m/s
16. a = 3 m/s²; distance = 37.5 m
17. a(before P) = 4 m/s², a(after P) = 6 m/s²; greater after P
18. a = 3 m/s²; x–t is a parabola
19. v(P–Q lead-in) = 5 m/s, v(after) = 1 m/s; first segment is faster
20. deceleration = 1.5 m/s²; distance = 27 m
21. Δv = 20 m/s; final velocity = 20 m/s
22. displacement = –40 m; motion is in the negative direction
23. a(A) = 6 m/s², a(C) = –6 m/s²; displacement = 126 m
24. velocity = 30 m/s; distance fallen = 45 m
25. v = –2 m/s; motion is in the negative x-direction
26. a = 3 m/s²; distance = 40 m
27. Δv (0–4 s) = 12 m/s; final velocity = 24 m/s
28. deceleration = 2.5 m/s²; distance = 5 m
29. x–t: straight slanted line; a–t: on the axis (zero)
30. velocity = 40 m/s; height = 80 m
31. a(A) = 8 m/s², a(B) = –4 m/s²; displacement = 48 m
32. net displacement = 40 m; total distance = 160 m
33. v(1st segment) = –3 m/s, v(2nd segment) = 3 m/s; net displacement = 3 m, distance = 21 m
34. time to top = 1.5 s; max height = 11.25 m
35. Δv (0–3 s) = 12 m/s; final velocity (t = 5 s) = 28 m/s
36. a = 3 m/s²; total distance = 138 m
37. displacement = 10 m; distance = 30 m
38. a = –6 m/s²; distance = 125 m